AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M348
Applied Statistical Modelling

Final Examination
Fall Semester 2025/2026

Date:19./Jan./2026

Number of Exam Pages: 7 (including this cover sheet)	Time Allowed: 3 hours

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1: True or False Questions [22 x 0.5=11 marks]

Which of the following statements are true/false?

1. Independence of residuals means that errors are not correlated with one another.
2. Outliers can have a large influence on regression coefficients.
3. A Q–Q plot is commonly used to assess whether residuals are normally distributed.
4. A small p-value provides strong evidence against the null hypothesis.
5. The t-test is used to test whether an individual regression coefficient is significantly different from zero.
6. The overall F-test and the individual t-tests always give identical conclusions about model significance.
7. The alternative hypothesis in regression testing states that all predictor variables have zero effect on the response.
8. Dummy variables are unnecessary when including categorical predictors in regression models.
9. In regression models with categorical variables, the intercept represents the mean response for all categories combined.
10. When regression lines for different groups are parallel, it implies that no interaction effect is present.
11. Logistic regression is a type of generalized linear model (GLM) used for binary outcome variables.
12. The canonical link function for the binomial family in GLMs is the log link.
13. Poisson regression is used when modeling data that represent counts.
14. The Gaussian family with an identity link function corresponds to the standard linear regression model.
15. An odds ratio greater than 1 in logistic regression indicates increased odds of success for a one-unit increase in the predictor.
16. In GLMs, deviance measures how well a model fits the data compared to a saturated model.
17. A larger deviance value indicates a better-fitting generalized linear model.
18. The null deviance in a GLM represents the deviance of a model that includes only an intercept.
19. In logistic regression, the link function is typically linear.
20. A less flexible model, such as linear regression, tends to perform better when the true relationship between predictors and response is linear.
21. Transforming variables can help satisfy the assumptions of a regression model.
22. Increasing model complexity tends to decrease the variance of the model's predictions.

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PART 2: Parallel slopes and non-parallel slopes models for the Life Expectancy dataset [24 marks]

ALL QUESTIONS ARE REQUIRED

The Life Expectancy Dataset contains data from 193 countries between 2000 and 2015, collected from WHO and UN sources to study factors influencing life expectancy. It includes 22 variables (two categorical: Country, Status; and 20 continuous predictors) covering immunization, mortality, economic, and social factors. In the parallel and non-parallel regression models, Life Expectancy is the response variable, while Status (Developed vs. Developing) is the factor, and GDP, BMI, and Alcohol consumption are continuous covariates — where the parallel model assumes equal slopes across groups, and the non-parallel model allows slopes to differ by Status. Variables used in the regression models:

- Life expectancy – Dependent variable representing the average expected lifespan in years.
- Status – Categorical factor indicating whether a country is Developed or Developing.
- GDP – Continuous covariate measuring the country's economic performance (per capita).
- BMI – Continuous covariate representing the population's average Body Mass Index, reflecting general health and nutrition.
- Alcohol – Continuous covariate representing the average annual alcohol consumption per capita, related to lifestyle and health behaviors.

1) Fit a parallel slopes regression model where the response variable is life_expectancy and the explanatory variables are status, gdp, bmi, and alcohol. Store the fitted model in an object called model_parallel. Use the summary () function to display the output of the model.

The resulting output of part (1) are given below:

```
Residuals:
    Min      1Q  Median      3Q     Max
-32.303  -4.154   0.919   4.295  22.338

Coefficients:
                    Estimate   Std. Error   t value   Pr(>|t|)
(Intercept)       6.492e+01  6.604e-01  98.300  <2e-16 ***
statusDeveloping -6.107e+00  5.080e-01 -12.021  <2e-16 ***
gdp               1.383e-04  1.175e-05  11.775  <2e-16 ***
bmi               2.148e-01  8.072e-03  26.607  <2e-16 ***
alcohol           3.613e-02  4.641e-02   0.778   0.436
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 7.044 on 2308 degrees of freedom
Multiple R-squared:  0.4747,    Adjusted R-squared:  0.4738
F-statistic: 521.5 on 4 and 2308 DF,  p-value: < 2.2e-1
```


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2) Explain what it means for this model to assume parallel slopes for the continuous covariates across different levels of status
3) Using the summary output, interpret the estimated value of the intercept in the context of the model.
4) What does the coefficient for statusDeveloping represent?
5) Interpret the estimated coefficient of gdp in this model. What does a positive value indicate?
6) Interpret the estimated coefficient of bmi.
7) Is there evidence that alcohol consumption has a statistically significant effect on life_expectancy? Justify your answer using the p-value.
8) Compare the Multiple R-squared and Adjusted R-squared. Why the adjusted R-squared is slightly lower?
9) What is the predicted life_expectancy for a developing country with gdp = 1000, bmi = 22, and alcohol = 5?
10) How would the predicted life expectancy change if gdp increased by 500, assuming other variables remain constant?
11) Why might we prefer a parallel slopes model over a non-parallel slopes model in some cases?
12) Suggest one potential limitation of assuming parallel slopes for this dataset.

Part B: Non-Parallel Slopes Model:
Fit a non-parallel slopes regression model where life_expectancy is the response variable, status is the factor, and gdp, bmi, and alcohol are continuous covariates. In this model, the slopes of continuous covariates are allowed to differ by status.

```
Residuals:
    Min      1Q  Median      3Q     Max
-32.287  -3.683   0.747   3.968  23.168

Coefficients:
                              Estimate Std. Error t value Pr(>|t|)
(Intercept)                  8.155e+01  1.477e+00  55.194  < 2e-16 ***
statusDeveloping            -2.404e+01  1.516e+00 -15.853  < 2e-16 ***
gdp                          6.263e-05  1.420e-05   4.412 1.07e-05 ***
bmi                         -1.051e-04  1.969e-02  -0.005 0.995742
alcohol                     -3.718e-01  1.162e-01  -3.200 0.001395 **
statusDeveloping:gdp         1.579e-04  2.361e-05   6.686 2.87e-11 ***
statusDeveloping:bmi         2.393e-01  2.153e-02  11.112  < 2e-16 ***
statusDeveloping:alcohol     4.347e-01  1.259e-01   3.454 0.000563 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.746 on 2305 degrees of freedom
Multiple R-squared:  0.5189,    Adjusted R-squared:  0.5174
F-statistic: 355.1 on 7 and 2305 DF,  p-value: < 2.2e-16
```


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13) Fit a non-parallel slopes regression model where interaction terms (status * gdp, status * bmi, status * alcohol) are included. Store the model in an object called model_nonparallel and display the summary.
14) Explain the difference between parallel and non-parallel slopes in terms of model assumptions.
15) Interpret the coefficient for statusDeveloping:gdp. What does this interaction term imply?
16) Interpret the coefficient for statusDeveloping:bmi.
17) Interpret the coefficient for statusDeveloping:alcohol.
18) Compare the residual standard errors and R-squared values of the parallel vs. non-parallel slopes models. Which model explains the variability of life_expectancy better?
19) Using the non-parallel slopes model, predict the life_expectancy for a developing country with gdp = 1000, bmi = 22, and alcohol = 5.
20) How does allowing non-parallel slopes improve model flexibility for this dataset?
21) What does the F-statistic and its p-value suggest about the overall significance of the non-parallel model?
22) If one continuous covariate had a non-significant interaction with status, would you consider removing it? Explain.

Part C: Assessing Predictor Importance Using ANOVA in Multiple Regression
We want to determine whether the predictor status should be included in the regression model for life expectancy. To do this, we first fit a reduced model excluding status and compare it to the full model using the anova() function. The ANOVA test evaluates whether including status significantly improves the model's ability to explain the variability in life_expectancy.

Comparison of full and reduced models:

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
2309	121687.9	NA	NA	NA	NA
2308	114517.8	1	7170.197	144.5087	2.487189e-32

23) How can you use the anova() function to test whether including an interaction term significantly improves the model?
24) What is the null hypothesis being tested when comparing reduced_model_status and full_model using ANOVA?
25) Based on the p-value in the ANOVA output, should status be included in the model? Explain your reasoning.
26) How does the change in Residual Sum of Squares (RSS) from the reduced to the full model reflect the contribution of status to predicting life_expectancy?
27) If the p-value had been larger than 0.05, how would your interpretation of the importance of status change?

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28) Write the R code to perform a similar ANOVA test to check whether alcohol should be included in the model.What does the p-value tell us about whether

PART 3: Log-linear models of two-way contingency tables data [15 x 1= 15 marks]

ALL QUESTIONS ARE REQUIRED

In this part, we will fit log-linear models of two-way contingency tables data. For that, we will use the data from the newborn births dataset which is presented as a two-way contingency table with:

- Two rows for the factor gender (one for the level male and one for female)
- Two columns for the factor induced (one for no and the other for yes)
- The counts of newborn babies from the sample (of 738 babies) in each of the four cells in the table.

1. Fit a log-linear model taking count to be the response, and taking gender and induced as explanatory variables. Store the results of the fitted model in the object modelFit, and display a summary of the results.

```
Deviance Residuals:
     1      2      3      4
0.4470 -0.8389 -0.5087  0.9159

Coefficients:
             Estimate Std. Error z value Pr(>|z|)
(Intercept)  5.52552    0.05896  93.716  < 2e-16 ***
gendermale   0.23962    0.07415   3.232  0.00123 **
inducedyes  -1.22167    0.08779 -13.916  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 243.8609  on 3  degrees of freedom
Residual deviance:   2.0012  on 1  degrees of freedom
AIC: 35.502

Number of Fisher Scoring iterations: 3
```

2. What type of model was fitted?
   a. Linear regression
   b. Logistic regression
   c. Poisson log-linear model
   d. ANOVA
3. What is the expected sign of the coefficient for inducedyes based on the data?
   a. Positive, since there are more induced than non-induced births
   b. Negative, since there are fewer induced births
   c. Zero, since induced and non-induced births are equal
   d. Not enough information

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4. Which of the following best describes the interpretation of the intercept?
   a. The expected log count for male babies with induced birth
   b. The expected log count for the reference group (female, non-induced)
   c. The total number of babies
   d. The difference in counts between male and female
5. Which factor combination is the reference category in the fitted model?
   a. Male, Yes
   b. Male, No
   c. Female, No
   d. Female, Yes
6. What does the z-value test in this summary output?
   a. Whether the residuals are normally distributed
   b. Whether the dispersion parameter equals 1
   c. Whether each coefficient is significantly different from zero
   d. Whether the AIC is minimal
7. What does a significant coefficient for gendermale mean?
   a. There is no difference between male and female counts.
   b. Gender is associated with different expected counts of births.
   c. Gender and induced are independent.
   d. The model is poorly fitted.
8. What is the total number of degrees of freedom for the null deviance?
   a. 1
   b. 2
   c. 3
   d. 4
9. Which statement best describes the meaning of the residual deviance being 2.0012 on 1 degree of freedom?
   a. The model fits poorly.
   b. The model fits well since residual deviance is small relative to df.
   c. The model is overfitted.
   d. The model explains less than half the variance.
10. The estimated mean count for female, non-induced births is:
   a. $e^{5.5255} \approx 250$
   b. $e^{0.2396} \approx 1.27$
   c. $e^{-1.2217} \approx 0.29$
   d. $e^{5.5255+0.2396} \approx 317$
11. Write down the fitted log-linear model equation using the output coefficients.
12. Based on the fitted model, what is the expected count for male, non-induced births?
13. Interpret the coefficient for inducedyes.
14. Interpret the coefficient for gendermale.
15. If we added an interaction term (gender:induced), what would it test?

End of questions

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